

COURSE DESCRIPTION

Course Number	0418-491	Course Title	Capstone Project I
Credit Hours	3 (3:0:3)	Course Coordinator	Capstone Project Committee
Compulsory	Yes	Pre/Corequisite	0418-390

Description

This course is taken by senior students under the guidance of a faculty member. The project proposal should be approved as per the department rules, policies, and guidelines. The project aims to provide students with a capstone design and development experience. It enables students to demonstrate and enhance their practical skills such as developing formal problem specifications, adapt to different computing environments and application domains, building prototypes, scientific writing, working in teams, interacting with users, and making oral presentations. Registered students must develop software preferably using a new environment, submit a comprehensive project report, and publicly present their work. The lecture component of the course explains the social and professional issues of computing as well as foundations of Human-Computer interactions.

Textbooks

Deborah G. Johnson (2009). *Computer Ethics (4th edition)*. Pearson.
ISBN-13: 9780131112414

Shneiderman, B., Plaisant, C., Cohen, M. S., Jacobs, S., Elmqvist, N., & Diakopoulos, N. (2016). *Designing the user interface: strategies for effective human-computer interaction (6th edition)*. Pearson.
ISBN-13: 978-0134380384

Major Topics Covered in the Course

	Topics	Hours
1	Basic principles of professional ethics and codes.	3
2	Ethical theories and concepts	6
3	Privacy and security principles	6
4	Digital intellectual property	6
5	Professional ethics	3
6	Usability of interactive systems, universal usability	3
7	Design, evaluation & user experience	6
8	Direct manipulation, fluid navigation, expressive languages	6
9	Exams	3
Total		42

Grading

Item	Weight	Provided by
Final Exam	40	Course Instructor
Proposal Presentation	10	Capstone Committee
Proposal Report	20	Capstone Committee
Project Prototype	30	Capstone Committee

Course Outcomes

Students should be able to:

1. Understand, explain, and analyze concepts and fundamentals of ethics and professionalism.
2. Demonstrate effective and professional collaboration.
3. Understand and explain HCI principles.
4. Produce software prototype and technical reports.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1			M	H		
2			M		H	
3	H					M
4	H	H			H	H

Last updated on Jan 8, 2024.

Approved by the capstone and curricula committee on Jan 8, 2024.